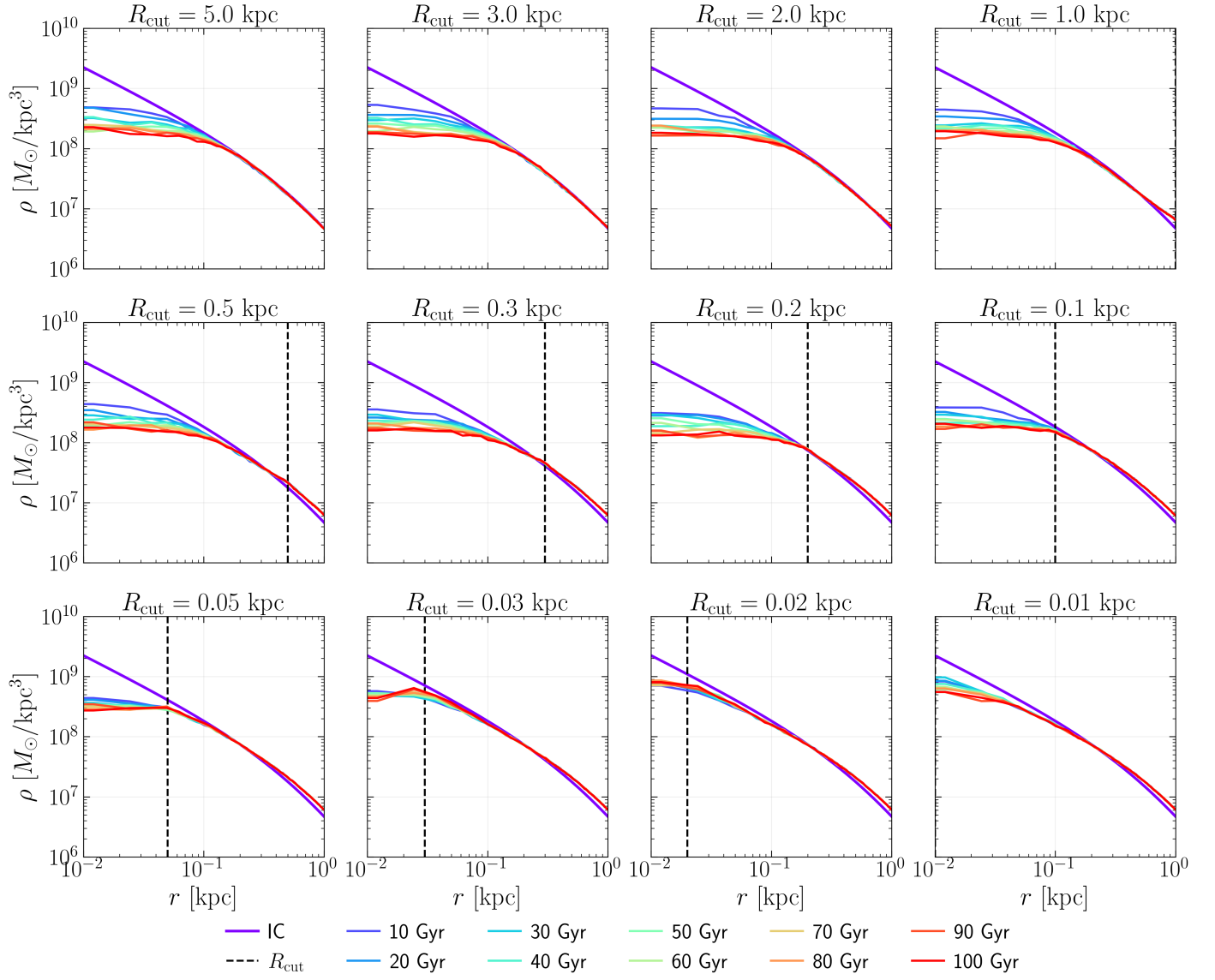




The figure above provides an overview of simulations with small R_{cut} values. For $R_{\text{cut}} \gtrsim 0.1 \text{ kpc}$, the system evolves very similarly to the reference (vanilla SWIFT) run. However, as R_{cut} approaches the softening length value, the cusp-to-core transition does not extend to large timesteps.

This behaviour is clearly illustrated in the following figures: while the simulation with $R_{\text{cut}} = 5 \text{ kpc}$ exhibits continued core growth up to 100 Gyr, progressively smaller values of R_{cut} lead to increasingly limited core evolution, eventually resulting in a density profile that does not decrease.

